FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES

CS054
(INTELLIGENT SYSTEMS)

EXERCISE
2

Sentiment Analysis

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I. PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design and evaluate solutions for computing problems. [PO: e]

II. COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- To design an Intelligent System based on methods learned. [CLO: 3]

III. INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Demonstrate understanding of Sentiment Analysis.
- Design and develop a simple program about Sentiment Analysis using Python.

IV. BACKGROUND INFORMATION

Sentiment Analysis can assist us in determining the mood and feelings of the general public as well as obtaining useful information about the setting. Sentiment Analysis is the process of assessing data and categorizing it according to the research's needs.

Sentiments can be used to gain a better understanding of different events and their consequences. It goes by various names, including "sentiment analysis," "opinion mining," "opinion extraction," "sentiment mining," "subjectivity analysis," "affect analysis," "emotion analysis," and "review mining," but they all serve the same objective. We might be able to figure out what people like or want by evaluating their feelings.



TextBlob

TextBlob is a python library and offers a simple API to access its methods and perform basic NLP tasks. Textblob's sentiment function returns polarity and subjectivity as two characteristics.

The polarity ranges from [-1,1], with 1 denoting a positive statement and -1 denoting a negative one. Subjective statements usually refer to personal feelings, emotions, or judgments, whereas objective sentences refer to facts. Subjectivity is also a float with a value between 0 and 1.

V. GRADING SYSTEM / RUBRIC (please see separate sheet)

VI. LABORATORY ACTIVITY

Perform the following:

1. Add the following code:

```
from textblob import TextBlob
```

2. Declare a variable that will accept a sentence:

```
sentence = input("Enter a sentence : ")
```

3. Next initialize TextBlob

```
s = TextBlob(sentence)
```

4. Then check the polarity of the sentence by using the code below:

```
p = s.sentiment.polarity
```

5. Make a decision based on the polarity. If the polarity of the sentence is less than zero the sentiment is negative. If the polarity is equals to zero, then the sentiment is neutral. If the polarity is greater than zero, the sentiment is positive.

```
#Negative p<0, Neutral p=0, Positive p>0
if p <0:
    print("Negative")
elif p==0:
    print("Neutral")
elif p>0 and p<=1:
    print("Positive")
```

6. Make the program perform the sentiment analysis 10 times and make a summary of how many sentences are positive, neutral or negative.

OUTPUT:

```
PS C:\Users\Joshua\Downloads\python\technicals> & "C:\Program Files\
sis.py
Enter sentence 1/10: I am Happy.
Positive
Enter sentence 2/10: I am Sad.
Negative
Enter sentence 3/10: You're handsome my friend!
Positive
Enter sentence 4/10: There are 3 computers.
Neutral
Enter sentence 5/10: I hate you!
Negative
Enter sentence 6/10: The movie was amazing!
Positive
Enter sentence 7/10: The meeting starts at noon.
Neutral
Enter sentence 8/10: This is excellent!
Positive
Enter sentence 9/10: nice catch!
Positive
Enter sentence 10/10: Why your so angry?
Negative

Summary of 10 sentences
Positive: 5
Neutral: 2
Negative: 3
PS C:\Users\Joshua\Downloads\python\technicals>
```

CODE:

```
from textblob import TextBlob

def main():
    positive = 0
    neutral = 0
    negative = 0

    for number in range(1, 11):
        sentence = input(f"Enter sentence {number}/10: ").strip()
        while not sentence:
            sentence = input("Please enter a non-empty sentence: ").strip()
```

```
        polarity = TextBlob(sentence).sentiment.polarity
        if polarity < 0:
            print("Negative")
            negative += 1
        elif polarity == 0:
            print("Neutral")
            neutral += 1
        else:
            print("Positive")
            positive += 1

    print("\nSummary of 10 sentences")
    print(f"Positive: {positive}")
    print(f"Neutral: {neutral}")
    print(f"Negative: {negative}")

if __name__ == "__main__":
    main()
```

CODE SCREENSHOT:

sentiment_analysis.py

Technicals > sentiment_analysis.py > ...

```python
from textblob import TextBlob


def main():
    positive = 0
    neutral = 0
    negative = 0

    for number in range(1, 11):
        sentence = input(f"Enter sentence {number}/10: ").strip()
        while not sentence:
            sentence = input("Please enter a non-empty sentence: ").strip()

        polarity = TextBlob(sentence).sentiment.polarity
        if polarity < 0:
            print("Negative")
            negative += 1
        elif polarity == 0:
            print("Neutral")
            neutral += 1
        else:
            print("Positive")
            positive += 1

    print("\nSummary of 10 sentences")
    print(f"Positive: {positive}")
    print(f"Neutral: {neutral}")
    print(f"Negative: {negative}")


if __name__ == "__main__":
    main()
```

VII. CHALLENGE

Write a program that will ask the user for a string containing a sentence. The program will then compute the happiness and sadness level of the sentence using the two functions described below. If the happiness level is higher than sadness level, then the tone of the sentence is happy. If the sadness level is higher, then the tone of the sentence is sad. Otherwise, it is neutral.

Find and print the tone of the sentence by printing the number of happy words followed by the number of sad words, followed by a simple statement of the analysis (whether positive, negative or neutral).

To accomplish this, you will write a function called number_happy(sentence) which returns the number of words in a given string called “sentence” that are happy. To do this, look for a public dataset containing a list of happy words. Find the total number of words in “sentence” that are in the happy words dataset.

Next, write a second function called number_sad(sentence) that works the same way but instead counts the number of sad words, also from a separate dataset.

OUTPUT:

```
PS C:\Users\Joshua\Downloads\python\technicals> & "C:\Program Files\
Enter a sentence: I am happy and delighted but sad.
Happy words: 2
Sad words: 1
Analysis: Positive (happy)
PS C:\Users\Joshua\Downloads\python\technicals>
```

```
PS C:\Users\Joshua\Downloads\python\technicals> & "C:\Program Files\
Enter a sentence: I feel sad and miserable.
Happy words: 0
Sad words: 2
Analysis: Negative (sad)
PS C:\Users\Joshua\Downloads\python\technicals>
```

```
PS C:\Users\Joshua\Downloads\python\technicals> & "C:\Program Files\
Enter a sentence: I am happy but sad.
Happy words: 1
Sad words: 1
Analysis: Neutral
PS C:\Users\Joshua\Downloads\python\technicals>
```

CODE:

```
import re
from pathlib import Path

DATA_FOLDER = Path(__file__).resolve().parent / "data"

def load_words(filename):
    # The dataset uses Latin-1; semicolon lines are comments.
    with (DATA_FOLDER / filename).open(encoding="latin-1") as file:
        return {line.strip().lower() for line in file
                if line.strip() and not line.startswith(";")}

HAPPY_WORDS = load_words("positive-words.txt")
SAD_WORDS = load_words("negative-words.txt")
```

```
def words_in(sentence):
    # Ignore case and punctuation; keep internal hyphens/apostrophes.
    return re.findall(r"[a-z0-9]+(?:[-'][a-z0-9]+)*(?:\+)?",
                      sentence.lower().replace("\u2019", "'"))

def number_happy(sentence):
    return sum(word in HAPPY_WORDS for word in words_in(sentence))

def number_sad(sentence):
    return sum(word in SAD_WORDS for word in words_in(sentence))

def main():
    sentence = input("Enter a sentence: ").strip()
    while not sentence:
        sentence = input("Please enter a non-empty sentence: ").strip()

    happy = number_happy(sentence)
    sad = number_sad(sentence)
    print(f"Happy words: {happy}")
    print(f"Sad words: {sad}")

    if happy > sad:
        print("Analysis: Positive (happy)")
    elif sad > happy:
        print("Analysis: Negative (sad)")
    else:
        print("Analysis: Neutral")

if __name__ == "__main__":
    main()
```

CODE SCREENSHOT:

challenge.py

Technicals > challenge.py > number_sad

```python
import re
from pathlib import Path

DATA_FOLDER = Path(__file__).resolve().parent / "data"


def load_words(filename):
    # The dataset uses Latin-1; semicolon lines are comments.
    with (DATA_FOLDER / filename).open(encoding="latin-1") as file:
        return {line.strip().lower() for line in file
                if line.strip() and not line.startswith(";")}


HAPPY_WORDS = load_words("positive-words.txt")
SAD_WORDS = load_words("negative-words.txt")


def words_in(sentence):
    # Ignore case and punctuation; keep internal hyphens/apostrophes.
    return re.findall(r"[a-z0-9]+(?:[-'][a-z0-9]+)*(?:\+)?",
                      sentence.lower().replace("\u2019", "'"))


def number_happy(sentence):
    return sum(word in HAPPY_WORDS for word in words_in(sentence))


def number_sad(sentence):
    return sum(word in SAD_WORDS for word in words_in(sentence))


def main():
    sentence = input("Enter a sentence: ").strip()
    while not sentence:
        sentence = input("Please enter a non-empty sentence: ").strip()

```

```
    happy = number_happy(sentence)
    sad = number_sad(sentence)
    print(f"Happy words: {happy}")
    print(f"Sad words: {sad}")

    if happy > sad:
        print("Analysis: Positive (happy)")
    elif sad > happy:
        print("Analysis: Negative (sad)")
    else:
        print("Analysis: Neutral")


if __name__ == "__main__":
    main()
```

VIII. REFERENCES

- Artasanchez A. Joshi P.(2020). Artificial Intelligence with Python: You Complete Guide to Building Intelligent apps using Python 3.x 2nd Edition. Packt Publishing
- Attaway, Stormy (2013). MATLAB ®. A Practical Introduction to Programming and Problem Solving Third Edition. Elsevier, Inc.

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